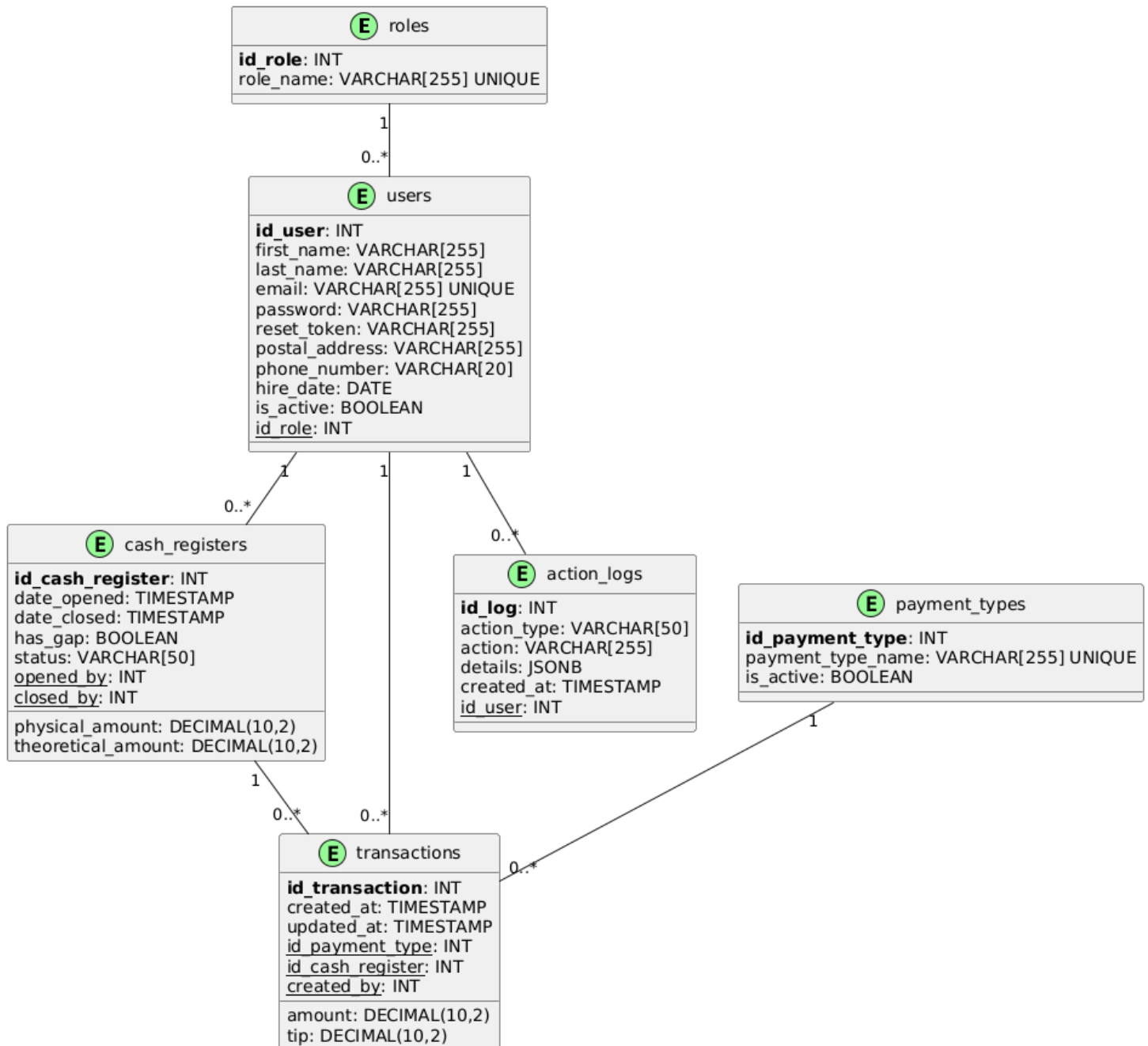


Document de conception - Application Web "Tour de Contrôle"

Diagramme ERD (Entité-Relation-Diagram)



@startuml

' Definition of entities

!define table(x) class x << (E,#98FB98) >>

!define primary_key(x) x

!define foreign_key(x) <u>x</u>

```
table(roles) {
    primary_key(id_role): INT
    role_name: VARCHAR[255] UNIQUE
}
```

```
table(users) {
    primary_key(id_user): INT
    first_name: VARCHAR[255]
    last_name: VARCHAR[255]
    email: VARCHAR[255] UNIQUE
    password: VARCHAR[255]
    reset_token: VARCHAR[255]
    postal_address: VARCHAR[255]
    phone_number: VARCHAR[20]
    hire_date: DATE
    is_active: BOOLEAN
    foreign_key(id_role): INT
}
```

```
table(payment_types) {
    primary_key(id_payment_type): INT
    payment_type_name: VARCHAR[255] UNIQUE
    is_active: BOOLEAN
}
```

```
table(cash_registers) {
    primary_key(id_cash_register): INT
    date_opened: TIMESTAMP
    date_closed: TIMESTAMP
    has_gap: BOOLEAN
    physical_amount: DECIMAL(10,2)
    theoretical_amount: DECIMAL(10,2)
    status: VARCHAR[50]
    foreign_key(opened_by): INT
    foreign_key(closed_by): INT
}
```

```
table(transactions) {
    primary_key(id_transaction): INT
    amount: DECIMAL(10,2)
```

```

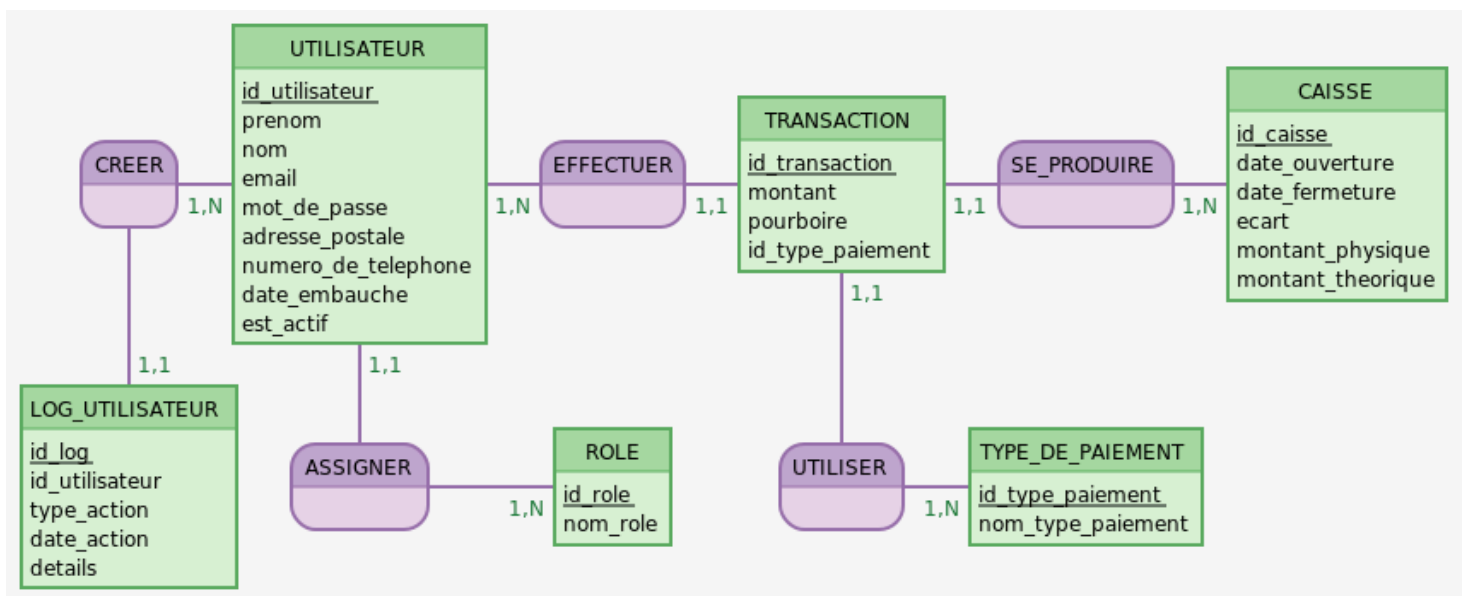
tip: DECIMAL(10,2)
created_at: TIMESTAMP
updated_at: TIMESTAMP
foreign_key(id_payment_type): INT
foreign_key(id_cash_register): INT
foreign_key(created_by): INT
}

table(action_logs) {
  primary_key(id_log): INT
  action_type: VARCHAR[50]
  action: VARCHAR[255]
  details: JSONB
  created_at: TIMESTAMP
  foreign_key(id_user): INT
}

roles "1" -- "0..*" users
users "1" -- "0..*" transactions
users "1" -- "0..*" cash_registers
users "1" -- "0..*" action_logs
payment_types "1" -- "0..*" transactions
cash_registers "1" -- "0..*" transactions
@enduml

```

MCD (Modèle Conceptuel de données)



%%mocodo

:::

UTILISATEUR: id_utilisateur, prenom, nom, email, mot_de_passe, adresse_postale, numero_de_teleph

:::

TRANSACTION: id_transaction, montant, pourboire, id_type_paiement 1, #id_utilisateur > UTILISATE

:::

CAISSE: id_caisse, date_ouverture, date_fermeture, ecart, montant_physique, montant_theorique

:

:

LOG_UTILISATEUR: id_log, id_utilisateur 1, type_action, date_action, details, #id_utilisateur 2

:::

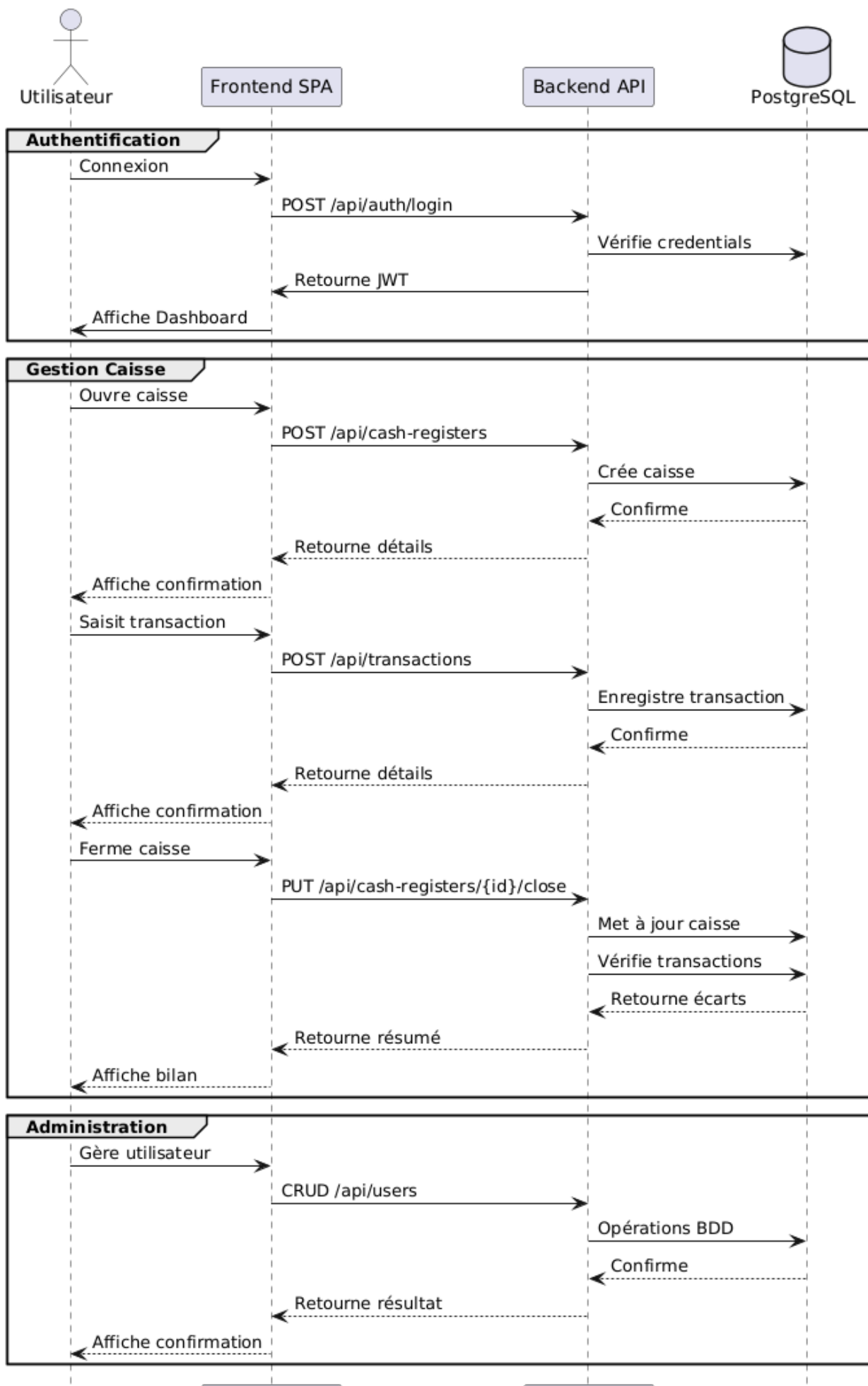
ROLE: id_role, nom_role

:::

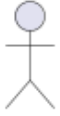
TYPE_DE_PAIEMENT: id_type_paiement, nom_type_paiement

:::

Diagramme de séquence



Utilisateur



Frontend SPA

Backend API

PostgreSQL



@startuml

' Acteurs

```
actor "Utilisateur" as USER
participant "Frontend SPA" as FRONT
participant "Backend API" as API
database "PostgreSQL" as DB
```

' Authentication

```
group Authentification
    USER -> FRONT: Connexion
    FRONT -> API: POST /api/auth/login
    API -> DB: Vérifie credentials
    API -> FRONT: Retourne JWT
    FRONT -> USER: Affiche Dashboard
end
```

' Cash Register

```
group Gestion Caisse
    USER -> FRONT: Ouvre caisse
    FRONT -> API: POST /api/cash-registers
    API -> DB: Crée caisse
    DB --> API: Confirme
    API --> FRONT: Retourne détails
    FRONT --> USER: Affiche confirmation

    USER -> FRONT: Saisit transaction
    FRONT -> API: POST /api/transactions
    API -> DB: Enregistre transaction
    DB --> API: Confirme
    API --> FRONT: Retourne détails
    FRONT --> USER: Affiche confirmation

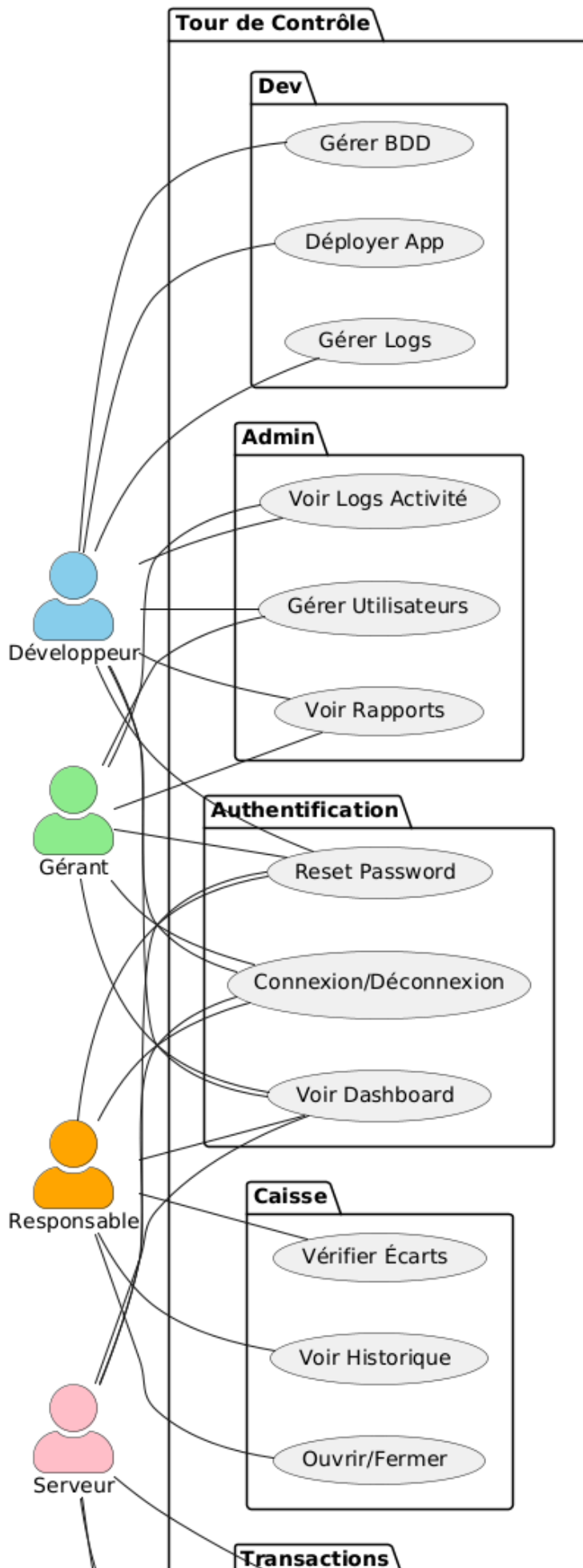
    USER -> FRONT: Ferme caisse
    FRONT -> API: PUT /api/cash-registers/{id}/close
    API -> DB: Met à jour caisse
    API -> DB: Vérifie transactions
    DB --> API: Retourne écarts
    API --> FRONT: Retourne résumé
    FRONT --> USER: Affiche bilan
end
```

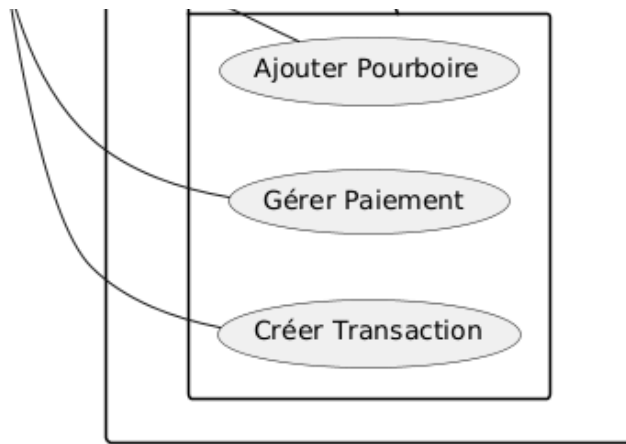
' Admin

```
group Administration
  USER -> FRONT: Gère utilisateur
  FRONT -> API: CRUD /api/users
  API -> DB: Opérations BDD
  DB --> API: Confirme
  API --> FRONT: Retourne résultat
  FRONT --> USER: Affiche confirmation
end
```

```
@enduml
```


Use Cases





@startuml

left to right direction

skinparam actorStyle awesome

' Acteurs

actor "Développeur" as D #SkyBlue

actor "Gérant" as G #LightGreen

actor "Responsable" as R #Orange

actor "Serveur" as S #Pink

package "Tour de Contrôle" {

' Use cases communs

package "Authentification" {

usecase "Connexion/Déconnexion" as UC1

usecase "Reset Password" as UC2

usecase "Voir Dashboard" as UC3

}

' Use cases développeur & gérant

package "Admin" {

usecase "Gérer Utilisateurs" as UC7

usecase "Voir Logs Activité" as UC8

usecase "Voir Rapports" as UC9

}

' Use cases développeur

package "Dev" {

usecase "Gérer Logs" as UC4

usecase "Déployer App" as UC5

usecase "Gérer BDD" as UC6

}

' Use cases responsable

package "Caisse" {

usecase "Ouvrir/Fermer" as UC10

usecase "Voir Historique" as UC11

```

    usecase "Vérifier Écarts" as UC12
}

' Use cases serveur
package "Transactions" {
    usecase "Créer Transaction" as UC13
    usecase "Gérer Paiement" as UC14
    usecase "Ajouter Pourboire" as UC15
}
}

' Relations communes
D -- UC1
D -- UC2
D -- UC3
G -- UC1
G -- UC2
G -- UC3
R -- UC1
R -- UC2
R -- UC3
S -- UC1
S -- UC2
S -- UC3

' Relations Admin (Dev & Gérant uniquement)
D -- UC7
D -- UC8
D -- UC9
G -- UC7
G -- UC8
G -- UC9

' Relations Dev uniquement
D -- UC4
D -- UC5
D -- UC6

' Relations Responsable
R -- UC10
R -- UC11
R -- UC12

' Relations Serveur
S -- UC13
S -- UC14
S -- UC15

```

Dictionnaire de donnée

Tables Principales

ROLES

Field	Type	Size	Description	Constraints
id_role	INT	-	Identifiant unique du rôle	PK, Auto-increment
role_name	VARCHAR	255	Nom du rôle	NOT NULL, UNIQUE

USERS

Field	Type	Size	Description	Constraints
id_user	INT	-	Identifiant unique	PK, Auto-increment
first_name	VARCHAR	255	Prénom	NOT NULL
last_name	VARCHAR	255	Nom	NOT NULL
email	VARCHAR	255	Email professionnel	NOT NULL, UNIQUE
password	VARCHAR	255	Mot de passe hashé	NOT NULL
reset_token	VARCHAR	255	Token réinitialisation	-
postal_address	VARCHAR	255	Adresse postale	-
phone_number	VARCHAR	20	Numéro de téléphone	-
hire_date	DATE	-	Date d'embauche	NOT NULL
is_active	BOOLEAN	-	Statut du compte	NOT NULL, Default true
id_role	INT	-	Rôle de l'utilisateur	FK -> ROLES

PAYMENT_TYPES

Field	Type	Size	Description	Constraints
id_payment_type	INT	-	Identifiant unique	PK, Auto-increment
payment_type_name	VARCHAR	255	Nom du type de paiement	NOT NULL, UNIQUE

Field	Type	Size	Description	Constraints
is_active	BOOLEAN	-	Statut d'activation	NOT NULL, Default true

CASH_REGISTERS

Field	Type	Size	Description	Constraints
id_cash_register	INT	-	Identifiant unique	PK, Auto-increment
date_opened	TIMESTAMP	-	Date d'ouverture	NOT NULL, Default NOW()
date_closed	TIMESTAMP	-	Date de fermeture	-
has_gap	BOOLEAN	-	Présence d'écart	Default false
physical_amount	DECIMAL	(10,2)	Montant physique	NOT NULL
theoretical_amount	DECIMAL	(10,2)	Montant théorique	NOT NULL
status	VARCHAR	50	État (OPEN/CLOSED)	CHECK IN ('OPEN','CLOSED')
opened_by	INT	-	Utilisateur ouvreure	FK -> USERS
closed_by	INT	-	Utilisateur fermeur	FK -> USERS

TRANSACTIONS

Field	Type	Size	Description	Constraints
id_transaction	INT	-	Identifiant unique	PK, Auto-increment
amount	DECIMAL	(10,2)	Montant	NOT NULL, CHECK > 0
tip	DECIMAL	(10,2)	Pourboire	Default 0, CHECK >= 0
created_at	TIMESTAMP	-	Date de création	NOT NULL, Default NOW()
updated_at	TIMESTAMP	-	Date de mise à jour	NOT NULL, Default NOW()
id_payment_type	INT	-	Type de paiement	FK -> PAYMENT_TYPES
id_cash_register	INT	-	Caisse associée	FK -> CASH_REGISTERS
created_by	INT	-	Utilisateur créateur	FK -> USERS

ACTION_LOGS

Field	Type	Size	Description	Constraints
id_log	INT	-	Identifiant unique	PK, Auto-increment
action_type	VARCHAR	50	Type d'action	CHECK IN ('AUTH','CASH','TRANSACTION','USER','SYSTEM')
action	VARCHAR	255	Description action	NOT NULL
details	JSONB	-	Détails additionnels	-
created_at	TIMESTAMP	-	Date de création	NOT NULL, Default NOW()
id_user	INT	-	Utilisateur associé	FK -> USERS

Notes

- Tous les champs TIMESTAMP incluent le fuseau horaire
- Les clés étrangères ont ON DELETE RESTRICT par défaut
- Des index sont créés sur les colonnes de jointure fréquentes